

Lecture Notes on Data Engineering
and Communications Technologies 46

Jennifer S. Raj
Abul Bashar
S. R. Jino Ramson *Editors*



Innovative Data Communication Technologies and Application

ICIDCA 2019



Springer

Lecture Notes on Data Engineering and Communications Technologies

Volume 46

Series Editor

Fatos Xhafa, Technical University of Catalonia, Barcelona, Spain

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Innovative Data Communication Technologies and Application

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ISSN 2367-4512 ISSN 2367-4520 (electronic)
Lecture Notes on Data Engineering and Communications Technologies
ISBN 978-3-030-38039-7 ISBN 978-3-030-38040-3 (eBook)
<https://doi.org/10.1007/978-3-030-38040-3>

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*We are honored to dedicate the proceedings
of ICIDCA 2019 to all the participants and
editors of ICIDCA 2019.*

Foreword

It is with deep satisfaction that I write this Foreword to the Proceedings of the ICIDCA 2019 held in RVS College of Engineering and Technology, Coimbatore, Tamil Nadu, India, on October 17–18, 2019.

This conference was bringing together researchers, academics and professionals from all over the world, and experts in Innovative Data Communication Technologies.

This conference particularly encouraged the interaction of research students and developing academics with the more established academic community in an informal setting to present and to discuss new and current work. The papers contributed the most recent scientific knowledge known in the field of distributed operating systems, middleware, databases, sensor, mesh, and ad hoc networks, quantum- and optics-based distributed algorithms, Internet applications, social networks, and recommendation systems. Their contributions helped to make the conference as outstanding as it has been. The Local Organizing Committee members and their helpers put much effort into ensuring the success of the day-to-day operation of the meeting.

We hope that this program will further stimulate research in theory, design, analysis, implementation, and application of distributed systems and networks. We feel honored and privileged to serve the best recent developments to you through this exciting program.

We thank all authors and participants for their contributions.

V. Gunaraj
Conference Chair

Preface

This Conference Proceedings volume contains the written versions of most of the contributions presented during the conference of ICIDCA 2019. The conference provided a setting for discussing recent developments in a wide variety of topics including dynamic, adaptive, and machine learning distributed algorithms, game-theoretic approaches to distributed computing, security in distributed computing, cryptographic protocols, fault tolerance, reliability, self-organization, self-stabilization, and so on. The conference has been a good opportunity for participants coming from various destinations to present and discuss topics in their respective research areas.

ICIDCA 2019 Conference tends to collect the latest research results and applications on Innovative Data Communication Technologies. It includes a selection of 95 papers from 274 papers submitted to the conference from universities and industries all over the world. All of the accepted papers were subjected to strict peer reviewing by 2–4 expert referees. The papers have been selected for this volume because of quality and relevance to the conference.

ICIDCA 2019 would like to express our sincere appreciation to all authors for their contributions to this book. We would like to extend our thanks to all the referees for their constructive comments on all papers, and especially, we would like to thank to organizing committee for their hard working. Finally, we would like to thank Springer publications for producing this volume.

V. Gunaraj
Conference Chair

Acknowledgments

ICIDCA 2019 would like to acknowledge the excellent work of our conference organizing the committee, keynote speakers for their presentation on October 17–18, 2019. The organizers also wish to acknowledge publicly the valuable services provided by the reviewers.

On behalf of the editors, organizers, authors, and readers of this conference, we wish to thank the keynote speakers and the reviewers for their time, hard work, and dedication to this conference. The organizers wish to acknowledge Dr. V. Gunaraj for the discussion, suggestion, and cooperation to organize the keynote speakers of this conference. The organizers also wish to acknowledge for speakers and participants who attend this conference. Many thanks are given to all persons who help and support this conference. ICIDCA would like to acknowledge the contribution made to the organization by its many volunteers. Members contribute their time, energy, and knowledge at a local, regional, and international level.

We also thank all the Chair Persons and Conference Committee Members for their support.

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A Secure Identity-Based Three-Party Authenticated Key Agreement Protocol Using Bilinear Pairings

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Abstract. The modern Internet technology era requires secure key agreement protocols, which are playing a significant role in the field of cryptography and network security. These protocols are mainly designed to establish a common session key between different parties. It can be easily found that various key agreement protocols are designed in the literature. However, many of these protocols are either proven insecure or have a burden of communication and computational cost. Therefore, a more secure key agreement protocol is needed. This paper exhibits an identity-based three-party authenticated key agreement (ID-3PAKA) protocol is devised, which securely and efficiently negotiates a common secret session key among three parties over the Internet. This protocol is based on the elliptic curve cryptography (ECC). It uses the idea of identity-based encryption (IBE) with bilinear pairings. The security of the proposed work is based on the hardness assumption of the discrete logarithm problem (DL) and elliptic curve. Further, we show that the proposed protocol ensures the known security properties of the session key.

Keywords: Key agreement · Elliptic curve · Hash function · Bilinear pairing · Forward secrecy

1 Introduction

The twentieth century grew with rapid development in the area of Internet and Mobile Communications Technologies, called Information and Communication Technology (ICT). ICT services have grown exponentially and have become

essential to the world in different ways. However, these technologies are changing very frequently, and several services with multimedia applications are growing through various real-life applications. In these regards, the security protection to various services becomes essential and challenging as well, and different security mechanisms for data encryption, authentication, integrity and availability are being developed at a rapid pace. In order to design different security mechanisms and meet the challenges, different cryptographic primitives are used in their convenient ways. The key agreement protocol is one of the significant tools of cryptography in which a secret session key is exchanged among a number of parties. In 1976, Diffie and Hellman [4] were the first who gave a new idea of having two separate keys; one for encipherment and other for decipherment. This protocol gave birth to the key agreement protocol, which is named as Diffie and Hellman (DH) key agreement protocol. Their proposal was used to transfer a common key among two authentic entities. Unfortunately, their protocol is vulnerable to a number of attacks including man-in-the-middle attack. To eliminate these difficulties, research has grown in this direction and many researchers have proposed different key agreement protocols [5, 14, 15]. Many of these protocols are not secure and suffer from different attacks. In this paper, we proposed a three-party authenticated key agreement (ID-3PAKA) protocol which is implemented in the framework of identity-based encryption (IBE) and uses bilinear pairings. The proof of correctness and security of our proposal is also presented to ensure the security of our ID-3PAKA protocol.

Cryptography is an art to use security primitives in a way to deal with the security challenges and meet the solutions. Data encryption is divided into two major categories namely, symmetric/private-key and asymmetric/public-key techniques in which the latter one has greater research impact than the former. However, the useful public-key cryptographic techniques like RSA, ElGamal, etc. have some disadvantage as they require extensive public key management overheads. Thus, a new technique called, IBE is introduced recently and is used by researchers to design efficient cryptographic tools for different security applications [6]. In this article, we formulated the idea of this technique to implement our protocol. Shamir [20] has firstly proposed the novel idea of IBE by choosing the known identity of a party which is supposed as his public-key. This known identity may be email id, phone number, physical address, etc. Using the identities, Shamir [20] removed the overhead of certificate management task from public-key cryptography. In addition, an entity called *private key generator* (PKG) is also considered to generate the identity-based private-key of the party. However, the practical IBE was implemented in 2001 by Boneh and Franklin [1]. Further, our proposed ID-3PAKA protocol using bilinear pairings. A bilinear pairing relates two members of a group to a member of another group. To implement our protocol, a bilinear mapping technique takes two members (points) of an elliptic curve group and maps it to a member of another multiplicative group. However, authentication to our proposed ID-3PAKA protocol is provided by means of the elliptic curve cryptography (ECC). The motivation to use the ECC in our protocol is that it provides better security than RSA, *i.e.*, a

160-bit key based on ECC exhibits the equal level of security which is provided by a 1024-bit key based on RSA.

1.1 Related Works

In 2004, Jeong et al. [16] designed many two-party key agreement protocols, which are executed in one round of communication. They claim that their proposed protocols are authenticated and resist known attacks including session key forward secrecy. In 2005, McCullagh and Barreto [18] devised a key agreement (ID-2PAKA) protocol for two parties using IBE. They showed that their protocol is efficient and secure than other existing state-of-the-art protocols. They also presented a comparative analysis for their protocol. Choo [2] reviewed McCullagh and Barreto's ID-2PAKA protocol and showed that their protocols are vulnerable if the attacker has sent the Reveal query. Hölbl et al. [12] devised an ID-2PAKA protocol using bilinear pairing. For their protocol, they used bilinear pairing and also derived a variant of signature scheme, which confirms the security of their protocol. They also showed that their proposal is comparatively secure and cost efficient than the related protocols. Gupta and Biswas [9] proposed two secure bi-partite key agreement protocols using IBE and bilinear pairing. The first protocol is based on the DH protocol; however the later is proposed for elliptic curve. They also proposed secure three-party key agreement protocols which are the extension of their bi-partite scheme. They showed that their protocols are secure against many attacks and claimed that these protocols exhibit better security and efficiency than other similar protocols.

Tseng [21] proposed an ID-2PAKA protocol whose security uses the hard assumption of discrete logarithm problem (DLP). It was also claimed that the performance of their proposal was better than other competing protocols. This protocol is secure and resistance to many possible attacks. Hölbl et al. [11] devised two ID-3PAKA protocols which uses bilinear pairing. They claimed that their protocols are secure on the defined security parameters. Gupta and Biswas [7] proposed a key transfer protocol for secure cellular communication. They also proved that their protocol exhibits strong security. Chou et al. [3] proposed two identity-based key agreement protocols for a mobile environments. Their first protocol was implemented for two-parties and the second protocol was an extension to the two-party key agreement protocol to three-party protocol. They also showed that their protocols achieve strong security notions. Gupta and Biswas [8] proposed an authenticated key agreement protocols in IBE framework. They also analyzed the performance and the security of protocol formally in the random oracle model. He et al. [10] developed a provably secure ID-3PAKA protocol using the elliptic curve for mobile environments. Islam et al. [13] proposed an improved ID-3PAKA protocol using ECC and hash function. They also proved that their protocol exhibits strong security using the AVISPA software.

2 Technical Details

The definitions and notations used in this paper is considered in this section and explained below:

2.1 Elliptic Curve Cryptography (ECC)

Miller [19] firstly introduced the concept of elliptic curve cryptosystem. Later, Koblitz [17] also proposed a cryptosystem based on the elliptic curve.

A cryptographic elliptic curve of prime order p is denoted by E/F_p and is expressed as the following equation:

$$y^2 \bmod p \equiv (x^3 + ax + b) \bmod p$$

In the above equation, a and b are the elements of F_p and satisfy the relation $(4a^3 + 27b^2) \not\equiv 0 \bmod p$. All the points of elliptic curve forms an additive abelian/commutative group. The “*addition operation*” (+) on elliptic curve points is defined as follows:

Let the two point of an additive group E/F_p be L and T . The addition of points L and T can be calculated as per the following conditions:

- Assume that L and T are different point on the curve having different x and y coordinates and the line joining these two points cuts the curve on another point $-Y$, then the reflection of $-Y$ about x -axis gives $L+T$, *i.e.*, $Y = L + T$. Here, the points Y and $-Y$ are known as additive inverse to each other with respect to x -axis.
- Assume that L and T are equal, *i.e.*, $L = T$. In this case, the line joining these same points will be the tangent to the curve and let it cuts the curve on a point $-Y$, then the image of $-Y$ about x -axis gives the addition $L + T$, *i.e.*, $Y = L + L = 2L$ or $2T$.
- Let two points to be added are inverse to each other *i.e.*, the points on the curve are L and $-L$, thus the line joining these points cuts the elliptic curve at infinity called “*point of infinity*” and denoted by O . Thus the addition is $L + (-L) = O$. The *point of infinity* is the identity of elliptic curve.
- The scalar point multiplication on the elliptic curve can be calculated by adding the same point repeatedly *i.e.*, kL can be calculated by adding L to itself k times using the point addition rules.

2.2 Bilinear Pairing

The technique of bilinear mapping is mainly used in the IBE to pair two groups of same order. In recent years, bilinear pairing has played an essential role in designing many cryptographic primitives. Let there are two groups namely G_1 and G_2 both having the same order q , where q is large prime number. The group G_1 is considered as an additive elliptic curve group, whereas group G_2 is a multiplicative group of same order. A bilinear mapping is denoted by $\hat{e} : G_1 \times G_1 \longrightarrow G_2$, which maps two elements of G_1 to an element in G_2 and defined by these three properties:

- **Bilinearity:** $\hat{e}(lL, tT) = \hat{e}(L, T)^{lt}$ for all $L, T \in G_1$ and $l, t \in \mathbb{Z}_q^*$.
- **Non-degeneracy:** $\hat{e}(L, L) = 1_{g_2}$ for all $L \in G_1$ and 1_{g_2} signify the multiplicative identity element of G_2 .
- **Computability:** The bilinear mapping $\hat{e}(L, T)$ can be efficiently computed for $L, T \in G_1$ by a polynomial-time-bounded algorithm.

2.3 Identity-Based Encryption (IBE)

A traditional non-identity-based public-key protocol involves lots of computation cost due to the public-key certificate management task. Further, a public-key certificate is also needed to be verified by the party for the validation of the public key. To eliminate the overhead of certificate management tasks in the non-identity-based public-key protocol, as early as 1984, Shamir [20] firstly gave an idea about the identity-based cryptography (IBC) where an identity of the party is considered as the public-key. In IBE, the public-key can be any arbitrary string and the sender uses the identity (say ID) of the receiver as public-key for encryption of a message. For instance, when Alice sends an encrypted message to Bob, she encrypts her message using Bob's email address (public identity) "bob@domainname.com". Then, Bob can decrypt the message using a related identity-based private-key obtained securely from the private key generator (PKG) through proper authentication. This novel concept of Shamir's protocol initiates a paradigm shift in public-key cryptographic techniques. However, until the year 2001, it was an open challenge for the researchers to construct a fully functional and efficient IBE. Boneh and Franklin [1] firstly proposed a fully functional IBE from the Weil Pairing. The IBE scheme depends on a trusted authority, called PKG, who generates the system's parameters *param* and the identity-based private-key of a party. The identity-based private-key is transferred to the party via a secure channel. With the nice functionality that any party can perform the encryption of messages with no prior key distribution among the parties (sender and receiver), IBE gains lots of attention from industry, research and academic community. The four algorithms of IBE are as follows:

- **Setup:** This algorithm is executed by the PKG. The input to *Setup* algorithm is a security parameter t and the outputs are *param* and the master private-key of PKG.
- **Extract:** This algorithm is used to compute the identity-based private key of a party user and it is executed by PKG. The input to *Extract* algorithm is the identity *ID* of a party and the output is the identity-based private key of the party.
- **Encrypt:** This algorithm is executed by the sender of a plaintext m . The input to *Encrypt* algorithm is m and the identity of the receiver, and the output is the encrypted message c , called ciphertext.
- **Decrypt:** This algorithm is executed by the receiver for decryption of the ciphertext c . The input to *Decrypt* algorithm is c and the identity-based private key of the receiver, and the output is plaintext m .

2.4 Computational Problems

- **Elliptic Curve Discrete Logarithm Problem (ECDLP):** This hard problem states that if $L, T \in G_1$ and L and T are known where $T = rL$ for $r \in \mathbb{Z}_q^*$, then it is hard to find r .
- **Bilinear Diffie-Hellman Problem (BDHP):** This hard problem states that if $L \in G_1$ and L, xL, yL and zL are known where $x, y, z \in \mathbb{Z}_q^*$, then it is hard to find $\hat{e}(L, L)^{xyz}$.
- **Computational Diffie-Hellman Problem (CDHP):** This hard problem states that if $L \in G_1$ and L, xL and yL are known where $x, y \in \mathbb{Z}_q^*$, then it is hard to find xyL .

Table 1. Notations

Symbols	Denotation
t	Security parameter
q	A prime modular
$H_1(\cdot)$	A cryptographic hash, where $H_1 : \{0, 1\}^* \rightarrow G_1$
$H_2(\cdot)$	A cryptographic hash, where $H_2 : G_2 \rightarrow \{0, 1\}^q$
G_1	An additive cyclic group on E/F_q
G_2	A multiplicative group of order q
$\hat{e}$	A bilinear map, where $\hat{e} : G_1 \times G_1 \rightarrow G_2$
$\mathbb{Z}_q$	$\{1, 2, \dots, q-1\}$
$\mathbb{Z}_q^*$	$\{x : 1 \leq x \leq q-1 \text{ mod } \gcd(x, q) = 1\}$
P	Primitive root of G
P_i	Party, where $i = 1, 2, 3$
s	$s \in \mathbb{Z}_q^*$ is the private key of PKG
P_{pk}	Master public key of PKG, where $P_{pk} = s \cdot P$

3 Proposed ID-3PAKA Protocol

The proposed ID-3PAKA protocol is described in this section. Our ID-3PAKA protocol is developed using IBE framework. In this protocol, security of negotiated session key and authentication of the parties are ensured using the hardness assumptions on elliptic curve group and DLP. However, the computational as well as communication complexities are also reasonable. Suppose the three parties, who have participated in key negotiation are P_1 , P_2 and P_3 and wants to negotiate a common session key among them. Except P_1 , P_2 and P_3 , the PKG is also involved in the execution of our ID-3PAKA protocol. For sake of clarity, Table 1 lists the different notations and their descriptions. The phases of our protocol are described as following steps:

3.1 Setup Phase

It takes a security parameter t as input and generates the following system's parameters. PKG chooses:

- a t -bit prime number q and groups G_1 and G_2 of same order q . Here G_1 is considered as an additive elliptic curve group, whereas group G_2 is a multiplicative group.
- a bilinear pairing $\hat{e} : G_1 \times G_1 \longrightarrow G_2$.
- a primitive element $P \in G_1$ and another primitive element $g \in G_2$.
- two hash functions H_1 and H_2 defined as $H_1 : \{0, 1\}^* \longrightarrow G_1$ and $H_2 : G_2 \longrightarrow \{0, 1\}^q$.
- a master private-key $s \in \mathbb{Z}_q^*$ which is secret to the PKG and master public key $P_{pk} = sP$.

PKG publicly announces system parameters $param = \langle q, H_1, H_2, G_1, G_2, P, \hat{e}, P_{pk} \rangle$ and kept master private-key s as secret to itself.

3.2 Key Extraction Phase

It is also executed by PKG to extract the identity-based private key of the party P_i , $\{i = 1, 2, 3\}$. The PKG executes the steps given below:

- A party P_i requests the PKG with their identities ID_i , $\{i = 1, 2, 3\}$.
- PKG checks the authenticity of P_i and calculates their private keys as $P_{pri} = sQ_i$, $\{i = 1, 2, 3\}$. Here, $Q_i = H_1(ID_i)$.
- The public key of P_i is Q_i . It may be noticed that there is no need of public key certificate to verify the public key Q_i of P_i .

3.3 Key Agreement Phase

The parties P_1 , P_2 and P_3 are involved in our protocol to negotiate a common session key between them. The steps used to exchange the session key are as follows:

- Party P_1 initiates the protocol and does the following:
 - Chooses an integer $a \in \mathbb{Z}_q^*$ at random and calculates $u = g^a \bmod q$ and $X_1 = aP$.
 - Calculates $Y_1 = aQ_1$ and $\mu_1 = \hat{e}(P_{pr1}, X_1)$ using his/her private key P_{pr1} .
 - Computes $\sigma_1 = u \oplus H_2(\mu_1)$.
 - Sends $\langle Y_1, u, \sigma_1 \rangle$ to P_2 .
- After receiving $\langle Y_1, u, \sigma_1 \rangle$, P_2 responds in the following manner:
 - Computes $\gamma_2 = H_2(\hat{e}(Y_1, P_{pk}))$ and checks whether $u \stackrel{?}{=} \sigma_1 \oplus \gamma_2$. If it is true, picks an integer $b \in \mathbb{Z}_q^*$ at random and computes $X_2 = bP$, $v = g^b \bmod q$ and $v' = u^b \bmod q$.
 - Calculates $Y_2 = bQ_2$ and $\mu_2 = \hat{e}(P_{pr2}, X_2)$ using his private key P_{pr2} .
 - Computes $\sigma_2 = v \oplus H_2(\mu_2)$ and $\sigma'_2 = v' \oplus H_2(\mu_2)$.

- Sends $\langle Y_2, v, v', \sigma_2, \sigma'_2 \rangle$ to P_3 .
- After receiving $\langle Y_2, v, v', \sigma_2, \sigma'_2 \rangle$, P_3 does the following:
 - Computes $\gamma_3 = H_2(\hat{e}(Y_2, P_{pk}))$ and checks whether $v \stackrel{?}{=} \sigma_2 \oplus \gamma_3$ and $v' \stackrel{?}{=} \sigma'_2 \oplus \gamma_3$. If these verifications are true, picks an integer $c \in \mathbb{Z}_q^*$ at random and computes the session key $K_3 = (v')^c \bmod q = g^{abc} \bmod q$.
 - Computes $X_3 = cP$, $w = g^c \bmod q$ and $w' = v^c \bmod q$.
 - Calculates $Y_3 = cQ_3$ and $\mu_3 = \hat{e}(P_{pr3}, X_3)$ using his private key P_{pr3} .
 - Computes $\sigma_3 = w \oplus H_2(\mu_3)$ and $\sigma'_3 = w' \oplus H_2(\mu_3)$.
 - Sends $\langle Y_3, w, w', \sigma_3, \sigma'_3 \rangle$ to P_1 .
- After receiving $\langle Y_3, w, w', \sigma_3, \sigma'_3 \rangle$, P_1 does the following:
 - Computes $\gamma_1 = H_2(\hat{e}(Y_3, P_{pk}))$ and checks whether $w \stackrel{?}{=} \sigma_3 \oplus \gamma_1$ and $w' \stackrel{?}{=} \sigma'_3 \oplus \gamma_1$. If these verifications are true, then computes the session key $K_1 = (w')^a \bmod q = g^{abc} \bmod q$.
 - Computes $u' = w^a \bmod q$ and $\sigma'_1 = u' \oplus H_2(\mu_1)$.
 - Sends $\langle u', \sigma'_1 \rangle$ to P_2 .
- After receiving $\langle u', \sigma'_1 \rangle$, P_2 does the followings:
 - Checks whether $u' \stackrel{?}{=} \sigma'_1 \oplus \gamma_2$ and computes the session key $K_2 = (u')^b \bmod q = g^{abc} \bmod q$.

Thus, it may be easily seen that $K = K_1 = K_2 = K_3 = g^{abc} \bmod q$. Note that the proposed protocol executes in only one round of communication and the proof of the verification $u \stackrel{?}{=} \sigma_1 \oplus \gamma_2$ is given below.

$$\begin{aligned}
 \sigma_1 \oplus \gamma_2 &= u \oplus H_2(\mu_1) \oplus \gamma_2 \\
 &= u \oplus H_2(\hat{e}(P_{pr1}, X_1)) \oplus H_2(\hat{e}(Y_1, P_{pk})) \\
 &= u \oplus H_2(\hat{e}(sQ_1, aP)) \oplus H_2(\hat{e}(aQ_1, sP)) \\
 &= u \oplus H_2(\hat{e}(Q_1, P)^{as}) \oplus H_2(\hat{e}(Q_1, P)^{as}) \\
 &= u
 \end{aligned}$$

The key agreement phase of our protocol is given in the Fig. 1.

4 Security Analysis

This section comes with the security analysis of our ID-3PAKA protocol. Our protocol resists all possible security attacks. The different type of security attacks are discussed below and it has been proved that our ID-3PAKA protocol is secure against all these types of attacks.

1. **Man-in-the-middle (MITM) Attack:** Our ID-3PAKA protocol negotiates a common session key and the security of this key is provided using the signature σ_i , $\{i = 1, 2, 3\}$ of the party P_i . In our protocol, P_i generates the signature σ_i using his/her private key. If an adversary $\mathcal{A}$ tries to impersonate any of the party P_i , $\{i = 1, 2, 3\}$, he/she cannot generate a valid signature and thus it is difficult for $\mathcal{A}$ to impersonate P_i . Hence, our ID-3PAKA protocol is not vulnerable to MITM attack.

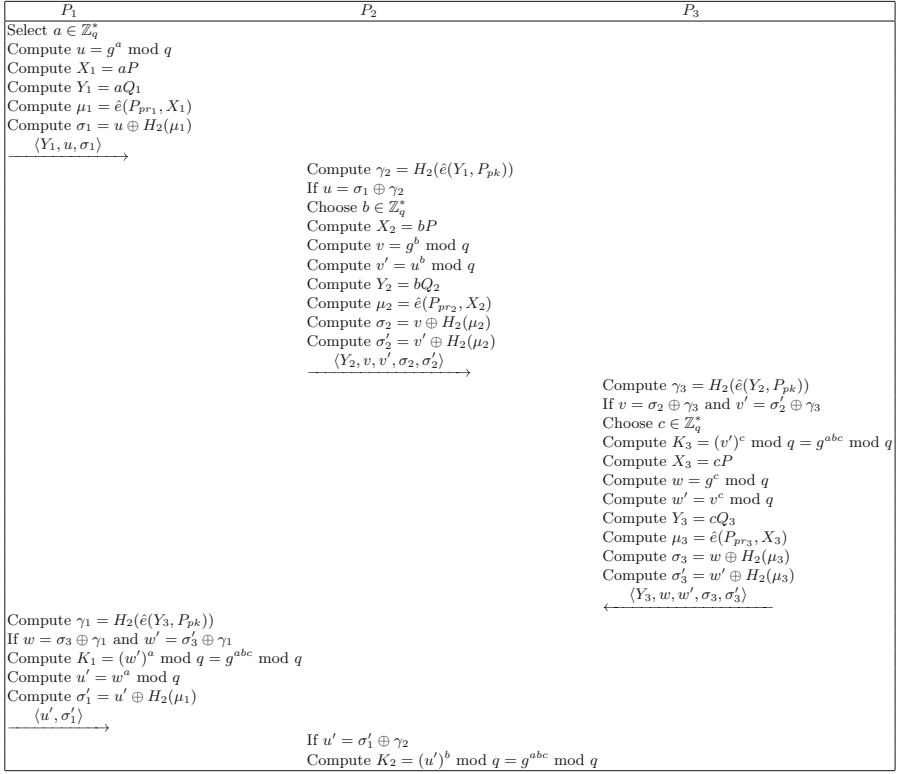


Fig. 1. Key agreement phase

2. **No key control (NKC):** This property ensures that the session key should not be a pre-selected value by any of the party or an adversary. A single party in our protocol cannot derive the actual session key because the ephemeral secrets a , b and c are not known to him/her. It is necessary for our protocol that each party P_i , $\{i = 1, 2, 3\}$ must be involved in the process of generating the session key. An individual party cannot enforce the correct key K to be a pre-selected value. Thus, our ID-3PAKA protocol ensures the NKC property of the session key.
3. **Known key security (K-KS):** This property ensures that an adversary cannot compute a current of any future session key even if any past session key is disclosed to him/her. The negotiated session key K in each session of the proposed protocol is unique because it is generated using the ephemeral secrets a , b and c chosen by P_1 , P_2 and P_3 at random. Thus, if an adversary $\mathcal{A}$ has access to previous session keys, he/she might not be able to derive the current or future session keys. Thus, our ID-3PAKA protocol ensures the K-KS property of the session key.
4. **Key-Compromise Impersonation (K-CI) Attack:** This property ensures that if the private-key of a party is disclosed to an adversary, however,

the adversary cannot impersonate other parties to the compromised party to compute the correct session key. Suppose, an adversary $\mathcal{A}$ somehow obtains the private-key, say P_{pr_1} of the party P_1 . Now, $\mathcal{A}$ tries to impersonate other parties P_2 and P_3 to P_1 , but he/she is not able to do so because he/she does not have the knowledge about the private-keys of P_2 and P_3 . Thus, the K-CI attack is prevented by our ID-3PAKA protocol.

5. **Perfect Forward Secrecy (PKS):** The PFS property ensures that an adversary cannot compute any past session key even if the private-keys of all parties are disclosed to him/her. An adversary $\mathcal{A}$ cannot recover the previous session keys even after getting the private-keys P_{pr_i} of the party P_i , $\{i = 1, 2, 3\}$ because in the process of key generation, the ephemeral secrets a , b and c are chosen at random. It is difficult for $\mathcal{A}$ to recover a , b and c from g^a , g^b and g^c , respectively because of the DLP. Thus, our ID-3PAKA protocol provides the PKS of the session key.
6. **Unknown Key-Share (UK-S) Attack:** The UK-S attack states that after completion of key agreement phase, the parties, say P_1 and P_2 believe that the session key is established with the party P_3 but, the P_3 feels that the session key is shared with an adversary $\mathcal{A}$. The proposed protocol requires the private keys P_{pr_i} and identities ID_i of parties P_i for $\{i = 1, 2, 3\}$ to generate the signature σ_i . An adversary $\mathcal{A}$ cannot generate the valid signature σ_i which can be verified by the other authentic parties. Hence, our proposed protocol resists UK-S attack.

5 Conclusion

The authors have devised a novel and efficient ID-3AKA protocol using IBE. The proposed protocol is executed for three parties and the negotiated session key is shared among these parties. For the agreement of the session key, we used the concept of ECC and bilinear pairing. The security of the session key, which is generated among three parties in each session, of our protocol is ensured using the hardness assumption of the DLP. Further, different attributes of the session key is ensured in our ID-3PAKA protocol.

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Machine Learning Based Stock Market Analysis: A Short Survey

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Abstract. Finance is one of the pioneering industries that started using Machine Learning (ML), a subset of Artificial Intelligence (AI) in the early 80s for market prediction. Since then, major firms and hedge funds have adopted machine learning for stock prediction, portfolio optimization, credit lending, stock betting, etc. In this paper, we survey all the different approaches of machine learning that can be incorporated in applied finance. The major motivation behind ML is to draw out the specifics from the available data from different sources and to forecast from it. Different machine learning algorithms have their abilities for predictions and are heavily depended on the number and quality of parameters as input features. This work attempts to provide an extensive and objective walkthrough in the direction of applicability of the machine learning algorithms for financial or stock market prediction.

Keywords: Stock market prediction · Machine Learning · Stock analysis

1 Introduction

The finance industry, in general, has been an early pioneer of AI technologies. Since the 70s, Wall Street has been using predictive models for predicting the prices of the market. In current era, the prime objective is to predict the market movements and how can these hedge funds be smartly invested at any given point of time based on market movements. Machine Learning techniques can boost the efficiency of an organization's operational accounting, financial reporting, allocations, adjustments, reconciliations, and intercompany transactions. The market for AI in finance is expected to grow to \$52.2 billion in 2021 [1]. The main purpose of machine learning in the financial market is to reduce the rate of false positives that a human hasn't anticipated while performing a stock transaction. ML techniques help us to detect the anomaly in a transaction data set.

1.1 Traditional Approach for Stock Market Prediction

There are different conventional approaches available which can help in stock prediction. Among all of these, two important approaches for stock market prediction are Efficient Market Hypothesis (EMH) and Random Walk theory. Efficient Market Hypothesis (EMH) theory states that asset prices fully reflect all available information. The simple explanation is that, it is not possible to perform better than the stock market on a risk-adjusted base as stocks should only respond to new information. There are three different forms in this hypothesis viz., (a) Weak EWH: only the past trading information is considered, (b) Semi-Strong EMH: all unbiased public information is utilized and (c) Strong EMH: all information public and private is used. Random Walk Theory states that stock market prices do not depend on past data and stock prices prediction is not possible due to great fluctuation.

The existence of price fluctuation prices in stock market, undiscovered serial correlations among stock prices, fundamental events and effect of economy are the great motivation behind adapting a better prediction model. This precise prediction can be the backbone for the traders and investors about the possibility of fluctuation in stock market [2]. Also, due to frequent data generation, real time stock prediction is one of the prime challenge. With the recent advancements in cloud computing architecture [3–6], machine learning module can be deployed on this high performance cloud.

1.2 Role of Machine Learning in Stock Market Prediction

Machine learning algorithms help to predict future outcomes based on available or past data. The prediction can be done based on regression and classification. Figure 1 depicts the various data sources, which are fed in to the Machine Learning model. The following subsection describes various machine learning approaches.

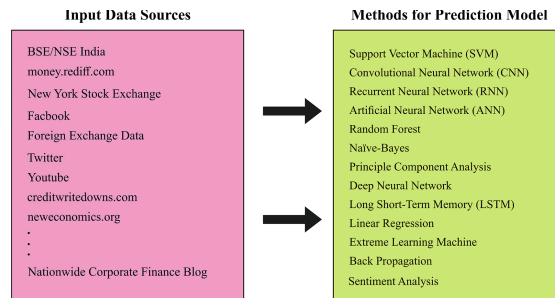


Fig. 1. Data sources and ML techniques for Stock Market Prediction

Multiple Regression Analysis: Regression is an ML technique which describes the relationship between a dependent variable and more than one independent variables. In stock market prediction, many factors are correlated to future stock prices. Regression analysis is one of the non-linear technique used for predicting the stocks. In this method, various market variables are analyzed and then the regression equation is formed using these variables. This equation is then utilized in the prediction model for predicting the dependent variable during the forecast period.